National University of Singapore
School of Computing
CS3244: Machine Learning
Solution to Tutorial 2 (2 weeks)

**The Linear Model**

**Contents:**
Tutorial 2A: Linear Regression

Tutorial 2B: Nonlinear Transforms, Gradient Descent

**Readings:** *Learning From Data*, Chapter 3.

For this tutorial, we prove that the error of linear regression model is bounded, especially to a specific value. Problem 1 proves some useful properties of the Hat Matrix which facilitates the proof of Problem 2.

1. *LFD*, Exercise 3.3 – **The Hat Matrix.**

   *The hat matrix is an integral part of understanding linear regression. We see the hat matrix makes an appearance in the analytical, closed form solution of linear regression, as it contains the pseudo inverse* $(\mathbf{X^T X})^{-1}\mathbf{X^T}$. *Let's look at its properties.*

   Consider the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X^T X})^{-1}\mathbf{X^T}$, where $\mathbf{X}$ is an $N$ by $d+1$ matrix, and $\mathbf{X^T X}$ is invertible.

   (a) Show that $\mathbf{H}$ is symmetric.
   *Show* $H^T = H$.

$$
\begin{aligned}
H^T &= (X(X^TX)^{-1}X^T)^T && \textit{Expand} && (1)\\
&= (X^T)^T(X(X^TX)^{-1})^T && \textit{Use } (AB)^T = B^TA^T && (2)\\
&= X((X^TX)^{-1})^TX^T && \textit{Use } (AB)^T = B^TA^T && (3)\\
&= X((X^TX)^T)^{-1})X^T && \textit{Use } (A^T)^{-1} = (A^{-1})^T && (4)\\
&= X(X^TX)^{-1}X^T. && && (5)\\
&= H. && \square && (6)
\end{aligned}
$$

   (b) Show that $\mathbf{H}^K = \mathbf{H}$, for any positive integer $K$.

   *We can prove this by induction. Note that the base case must involve matrix multiplication, so we cannot start from* $k = 1$, *but should start from* $k = 2$.

*<u>Base Case</u> ($k = 2$): Show $H^2 = H$.*

$$
\begin{aligned}
H^2 &= (X(X^TX)^{-1}X^T)(X(X^TX)^{-1}X^T) && \textit{Expand} && (1)\\
&= (X\underbrace{(X^TX)^{-1}}_{A}\underbrace{(X^TX)}_{B}(X^TX)^{-1}X^T) && \textit{Associate} && (2)\\
&= (X(X^TX)^{-1}X^T) && \textit{Cancel } A \text{ \& } B \textit{ to } I,\ \textit{drop } I && (3)\\
&= H. && \square && (4)
\end{aligned}
$$

*<u>Inductive Step</u>: Given $k$ is true, show for $k+1$. Show $H^{k+1} = H$.*

$$
\begin{aligned}
H^{k+1} &= (H^k)(X(X^TX)^{-1}X^T) && \textit{Expand} && (5)\\
H^{k+1} &= (H)(X(X^TX)^{-1}X^T) && \textit{Premise} && (6)\\
&= (X\underbrace{(X^TX)^{-1}}_{A}\underbrace{(X^TX)}_{B}(X^TX)^{-1}X^T) && \textit{Associate} && (7)\\
&= (X(X^TX)^{-1}X^T) && \textit{Cancel } A \text{ \& } B \textit{ to } I,\ \textit{drop } I && (8)\\
&= H. && \square && (9)
\end{aligned}
$$

*This property $X = X^k$ is called* **idempotency**. *In fact, orthogonal projection matrices (which* **H** *is a member of), exactly have both the symmetric and idempotent properties (i.e., "if and only if")*

(c) If **I** is the identity matrix of size $N$, show that $(\mathbf{I}-\mathbf{H})^K = \mathbf{I}-\mathbf{H}$ for any positive integer $K$.

*We can prove this similar to Part (b), using an inductive proof. We just detail the core part here.*

$$
\begin{aligned}
(I-H)^2 &= (I-H)(I-H) && \textit{Expand} && (1)\\
&= I^2 - 2IH + H^2 && \textit{Multiply out} && (2)\\
&= I - 2IH + H && I \textit{ and } H'\textit{s idempotency; Part (b)} && (3)\\
&= I - H. && \textit{Drop } I, \textit{associate } \square && (4)
\end{aligned}
$$

*You might guess that $(I - H)$ also is an orthogonal projection matrix, since it is also symmetric and idempotent. And you'd be right.*

(d) (Optional) Show that $trace(H) = d + 1$, where the *trace* ($tr$) is the sum of diagonal elements. [*Hint:* $trace(\mathrm{AB}) = trace(\mathrm{BA})$.]

$$
\begin{aligned}
trace(H) &= trace(\underbrace{X}_{\text{A}}\underbrace{(X^TX)^{-1}X^T}_{\text{B}}) && \textit{Expand} && (1)\\
&= trace((X^TX)^{-1}X^TX) && \textit{Apply hint} && (2)\\
&= trace(I_{(d+1)}). && \underbrace{X^{-1}}_{\text{N}\times(d+1)}\underbrace{X}_{(d+1)\times N} = I\ \square && (3)
\end{aligned}
$$

*For a symmetric and idempotent matrix $A$, $rank(A) = trace(A)$. We'll use these properties in the next question.*

2. *LFD*, Exercise 3.4 – **Inducing Noise**.

   Consider a noisy target $y = \mathbf{w}^{*T}\mathbf{x} + \epsilon$ for generating the data, where $\epsilon$ is a noise term with zero mean and $\sigma^2$ variance, independently generated for every samples $(\mathbf{x}, y)$. The expected error of the best possible linear fit to this target is thus $\sigma^2$.

   For the data $\mathcal{D} = (\mathbf{x}_1, y_1), \cdots, (\mathbf{x}_N, y_N)$, denote the noise in $y_n$, as $\epsilon_n$ and let $\epsilon = [\epsilon_1, \epsilon_2, \cdots, \epsilon_N]^T$; assume that $X^TX$ is invertible. By following the steps below, show that the expected in-sample error of linear regression with respect to $\mathcal{D}$ is given by:

$$\mathbb{E}[E_{in}(\mathbf{w}_{lin})] = \sigma^2(1 - \tfrac{d+1}{N})$$

   (a) Show that the in-sample estimate of $\mathbf{y}$ is given by $\hat{\mathbf{y}} = X\mathbf{w}^* + H\epsilon$.

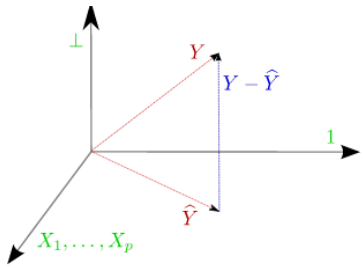
   *Note that in the first line of the description of this problem, the noisy target $y = \mathbf{w}^{*T}\mathbf{x}+\epsilon$ a refers to a single data instance. This question asks about $\mathbf{y}$, the vector of all of the observed outputs (class labels).*

   *For a sample instance, we have $y = \mathbf{w}^{*T}\mathbf{x} + \epsilon$. By "stacking" all $N$ of the examples on top of each other into a matrix form (cf Slide 19, Week 2), we obtain:* $\mathrm{y} = Xw^* + \epsilon$ *(nb, here $\mathbf{y}$ and $\epsilon$ are vectors of $N$ outputs and error values, one for each point, respectively). To obtain the predicted values, $\hat{\mathbf{y}}$, we multiply through by $H$; i.e., $\hat{\mathbf{y}} = H\mathbf{y} = H(Xw^*+\epsilon)$. Expanding, we obtain $\hat{\mathbf{y}} = X\mathbf{w}^* + H\epsilon$.*

   *<u>Aside.</u> The hat matrix gets its name from this property. It is the matrix that we multiply the original outputs $\mathbf{y}$ to obtain the predicted values $\hat{\mathbf{y}}$; i.e., $\hat{\mathbf{y}} = H\mathbf{y}$. We can say $H$ maps a column vector $\mathbf{y}$ into the space spanned by $X$.*

   *This is easier to understand graphically. In Figure 1, the linear regression hyperplane is represented by the plane at the "bottom" defined by $X$. Let's look at a particular instance. $y$ is the observed output, here a bit above the plane. $Hy = \hat{y}$ is thus the point on the linear regression plane that is the projection of $y$ onto $X$. It is an* **orthogonal** *projection, as $\hat{y}$ is the point that represents the shortest distance to $y$.*

   **Figure 1.** Linear regression as projection. Taken from `http://web.stanford.edu/class/stats191/notebooks/figs/axes_multiple_full.svg`.




(b) Show that the in-sample error vector $\hat{\mathbf{y}} - \mathbf{y}$ can be expressed by a matrix times $\epsilon$. What is the matrix?

*From Part (a), we have:*

$$\begin{aligned} \hat{y} - y &= Xw^* + H\epsilon - (Xw^* + \epsilon) && \textit{Expand} && (1) \\ &= (H - I)\epsilon. && \textit{Cancel and factor } \square && (2) \end{aligned}$$

*Related, the matrix $M = (I - H)$ (cf Problem 1 (c)) is the orthogonal projection matrix that projects $y$ into its residual $\hat{y} - y$. This is the vertical line marked in blue, in Figure 1. You can see that $y = Hy + My$.*

(c) Express $E_{in}(\mathbf{w_{lin}})$in terms of $\epsilon$ using (b), and simplify the expression using Problem 1 (c).

$E_{in}(\mathbf{w}_{lin}) = \frac{1}{N}(\hat{\mathbf{y}} - \mathrm{y})^T(\hat{\mathbf{y}} - \mathrm{y}) = \frac{1}{N}(\epsilon^T(H - I)^T(H - I)\epsilon)$.

*It can be seen that (H-I) is also symmetric, also according to Problem 1 (b). Expanding the above equation and rearranging, we have:*

$$E_{in}(\mathbf{w}_{lin}) = \frac{1}{N}(\epsilon^T(I - H)\epsilon)$$

(d) Prove that $\mathbb{E}[E_{in}(\mathbf{w}_{lin})] = \sigma^2(1 - \frac{d+1}{N})$ using (c) and the independence of $\epsilon_1, \cdots, \epsilon_N$. [ *Hint: The sum of diagonal elements of a matrix (the trace) will play a role. See Problem 1 (d).* ]

*Continuing from (c), we have*

$$E_{in}(\mathbf{w}_{lin}) = \underbrace{\frac{1}{N}\epsilon^T\epsilon}_{left} - \underbrace{\frac{1}{N}\epsilon^T H\epsilon}_{right}$$

*Here, we can see for the left part:*

$$\mathbb{E}[\frac{1}{N}\epsilon^T\epsilon] = \frac{1}{N}\mathbb{E}(\epsilon^2) = \frac{1}{N}(var(\epsilon) + \mathbb{E}(\epsilon)^2) = \frac{N\sigma^2}{N} = \sigma^2$$

*and for the right part, we have:*

$$\frac{1}{N}\epsilon^T H\epsilon = \frac{1}{N}\{\sum H_{ii}\epsilon_i^2 + \sum_{i\neq j} H_{ij}\epsilon_i\epsilon_j\}$$

*and for left part of this above equation,*

$$\mathbb{E}[\sum H_{ii}\epsilon_i^2] = \sum H_{ii}\mathbb{E}[\epsilon_i^2] = trace(H)\sigma^2 = (d+1)\sigma^2$$

*We also note since $\epsilon_i$ and $\epsilon_j$ are independent:*

$$\mathbb{E}[\sum_{i\neq j} H_{ij}\epsilon_i\epsilon_j] = \mathbb{E}[\sum H_{ii}]\mathbb{E}[\epsilon_i]\mathbb{E}[\epsilon_j] = 0$$

*rearranging the above equations, we get:*

$$\mathbb{E}_D[E_{in}(\mathbf{w}_{lin})] = \sigma^2(1 - \frac{d+1}{N})$$

For the expected out-of-sample error, we take a special case which is easy to analyze. Consider a test data set $\mathcal{D}_{test} = (\mathbf{x_1}, y_1^{'}), \cdots, (\mathbf{x_N}, y_N^{'})$, which shares the same input vectors $\mathbf{x}_n$ with $\mathcal{D}$ but with a different realization of the noise terms. Denote the noise in $y_n^{'}$ as $\epsilon_n^{'}$ and let $\epsilon_n^{'} = [\epsilon_1^{'}, \cdots, \epsilon_N^{'}]^T$. Define $E_{test}(\mathbf{w}_{lin})$ to be the average squared error on $\mathcal{D}_{test}$.

(e) Prove that $\mathbb{E}[E_{test}(\mathbf{w}_{lin})] = \sigma^2(1 + \frac{d+1}{N})$.

*The proof is similarly developed like in (d). We have:*

$$\hat{\mathbf{y}} - \mathbf{y} = H\epsilon - I\epsilon^{'}$$

*Therefore, out of sample error $E_{test}$:*

$$E_{test}(w_{lin}) = \frac{1}{N}\|\hat{y} - y\| = \frac{1}{N}(H\epsilon - Ie^{'})^T(H\epsilon - Ie^{'}) = \frac{1}{N}\{\epsilon^T H\epsilon - \epsilon^T H\epsilon^{'} - \epsilon^{'T} H\epsilon + \epsilon^{'T}\epsilon^{'}\}$$

*from (d) we have $\mathbb{E}[\epsilon^T H\epsilon] = \sigma^2(d+1)$ and $\mathbb{E}[\epsilon^{'T}\epsilon^{'}] = N\sigma^2$*
*Since $\epsilon^{'}$ and $\epsilon$ are i.i.d., $\mathbb{E}[\epsilon^T H\epsilon'] = 0$ and $\mathbb{E}[\epsilon^{'T} H\epsilon] = 0$. Rearranging the above equations, we prove:*

$$\mathbb{E}[E_{test}(\mathbf{w}_{lin})] = \sigma^2(1 + \frac{d+1}{N})$$

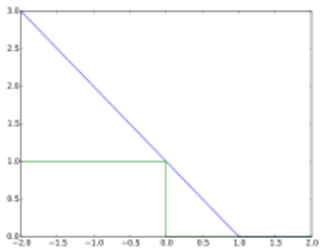
*We have thus proved that the in-sample and out-of-sample error of linear regression is bounded to a specific value (given simple assumption of the noise item as a Gaussian).*

3. **Error bound**. In classification problems, we have $err_{in} = [sign(\mathbf{w}^T\mathbf{x}) \neq y]$ for $y \in \{-1, +1\}$. Which of below is/are the upper bound of $err_{in}$? Explain your answer.

(a) $err(\mathbf{w}) = max(0, 1 - y\mathbf{w}^T x)$.

*True, it is an upper bound to the true loss 0/1 function.*

*When* $sign(\mathbf{w}^T x = sign(y), error_{in} = 0$, $err(\mathbf{w}) \geq 0$; *when* $sign(\mathbf{w}^T x \neq sign(y)$, $-y\mathbf{w}^T x > 0$ *and so that* $1 - y\mathbf{w}^T x > 1 = error_{in}$. *This is the* **Hinge Loss** *function, commonly used in the training of support vector machines.*



*Hinge Loss (blue) versus 0/1 loss (green). Taken from* https://en.wikipedia.org/wiki/Hinge_loss.

(b) $err(\mathbf{w}) = max(0, 1 - y\mathbf{w}^T x)^2$.

*True, as can be seen from (a).*

(c) $err(\mathbf{w}) = \theta(-y\mathbf{w}^T x)$, where $\theta(s) = \frac{exp(s)}{1+exp(s)}$.

*False. Under a single dimension space, we can be seen from the plot of sigmoid function that* $\theta(s) < 1$.

(d) $err(\mathbf{w}) = max(0, -y\mathbf{w}^T x)$.

*False. Consider when* $sign(\mathbf{w}^T x \neq sign(y)$ *and uni-dimension situation. One can draw the plot of* $f(w) = -ywx$, *which is the "hinge function" as well. When* $1 < -y\mathbf{w}x < 0$, $f(w) < 1$.

(e) $err(\mathbf{w}) = exp(-y\mathbf{w}^T x)$.

*True. This can be seen from the plot of* $f(s) = exp(s)$. When $y\mathbf{w}x > 0$, $0 < err(\mathbf{w}) < 1$, when $y\mathbf{w}x < 0$, $err(\mathbf{w}) > 1$

4. **Nonlinear Transformations.** Consider the feature transform $\Phi(1, x_1, x_2) = (1, x_1^2, x_2^2)$. Draw and show the boundary (not strictly) in $\mathcal{X}$ that a hyperplane $\hat{\mathbf{w}}$ in $\mathcal{Z}$ correspond to under the following cases:
(You can find the draft of the plot from webcast slides.)

(a) $\hat{w}_1 > 0, \hat{w}_2 < 0$.

*When $\hat{w}_0 > 0$, the plot shows a "North-South opening hyperbola"; when $\hat{w}_0 < 0$, it's a "East-West opening hyperbola".*

(b) $\hat{w}_1 > 0, \hat{w}_2 = 0$.

*When $\hat{w}_0 > 0$, there's no border in $\mathcal{X}$ space, all data points $\mathbf{x}$ are classified into $y = +1$ category. When $\hat{w}_0 < 0$, the borders are two symmetric lines that perpendicular to $x_1$ axis.*

(c) $\hat{w}_1 > 0, \hat{w}_2 > 0, \hat{w}_0 < 0$.

*It's either a ellipse or a circle, which is similar to Figure 3.5(a) of the textbook, whereas the classification is reversed.*

(d) $\hat{w}_1 > 0, \hat{w}_2 > 0, \hat{w}_0 > 0$.

*There's no border in $\mathcal{X}$ space. All data points $\mathbf{x}$ are classified into $y = +1$ category as well.*

5. **Stochastic Gradient Descent.** Let the error for a single data point $(\mathbf{x}_n, y_n)$ be defined as $e_n(\mathbf{w}) = \max(0, -y_n \mathbf{w}^t \mathbf{x}_n)$.

(a) Plot $e_n(\mathbf{w})$ for situation where $\mathbf{x}_n$ is univariate.
*When $-y_n x_n > 0$, the plot is Figure 2(a), when $-y_n x_n < 0$, the plot is Figure 1 (b).*

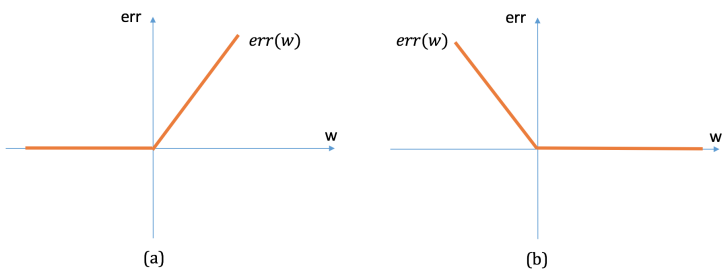



Figure 2: the plot for error function

(b) Show the batch gradient descent formulation of the error function (you may ignore notion of a break point).

*As can be seen from (a), $err(\mathbf{w})$ is continuous but not differentiable over all possible $\mathbf{w}$. Since we cannot directly compute the gradient of $err(\mathbf{w})$ over all $\mathbf{w}$, we can compute the subgradient of err at break point. (**Hint**: the subgradiant of a function refers to any plane/line that lower bounds the function at the given point)*

*Then we compute the subgradient of err:*

*i. When* $y\mathbf{wx} > 0$, $\nabla err = 0$

*ii. When* $y\mathbf{wx} < 0$, $\nabla err = -y\mathbf{x}$

*iii. When* $y\mathbf{wx} = 0$, $\nabla err \in [-y\mathbf{x}, 0]$ *E.g. -*$y\mathbf{x}$*.*

*Putting above in one line, we have:*

$$\nabla err(\mathbf{w}) = \sum_{i:y\mathbf{wx_i}<0} -y_i\mathbf{x}_i$$

(c) Argue that PLA is essentially a specific form as SGD on $e_n$.

*For the case which only considers t that lets* $y_t\mathbf{wx}_t < 0$*, the gradient of* $err(\mathbf{w}_t) = -y_t\mathbf{x_t}$*. Therefore, among those misclassified points, we randomly pick one of them.*

*Then the updating direction become*

$$\mathbf{v}_t = -\nabla err(\mathbf{w}_t) = y_t\mathbf{x}_t$$

*And we have the weight updating function:*

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \eta\mathbf{v_t} = \mathbf{w}_t + \eta y_t\mathbf{x}_t$$

*When* $\eta = 1$*, the above equation is the updating equation in perceptron algorithm.(Refering to (1.3) at LFD, page 7)*

6. **Linear Regression and Maximum Likelihood Estimation.** Consider a linear regression of the probabilistic form, which means $y = \mathbf{w^T x} + \epsilon_\mathbf{t}$, where the noisy item $\epsilon_\mathbf{t} \sim \mathcal{N}(0, \sigma^2)$ and is iid (identically independently distributed) for each t.

(a) Show the likelihood function of the model.

*Since* $Y_i = X_i\mathbf{w} + \epsilon_i$*, and we have* $\epsilon \sim \mathcal{N}(0, \sigma^2)$*, according to the property of gaussian variables, we have* $Y_i \sim \mathcal{N}(X_i\mathbf{w}, \sigma^2)$*. Therefore, the p.d.f of random variable* $Y_i$ *is*

$$p(Y_i|\sigma^2, X) = \frac{1}{\sqrt{2\pi}\sigma} exp\{\frac{-1}{2\sigma^2}(y_i - X_i\mathbf{w})^2\}$$

*Then with sequence* $Y = Y_1, Y_2, \cdots, Y_n$*, the likelihood function in terms of* $Y$ *is*

$$\mathcal{L} = \prod_{i=1}^{N} p(Y_i|\sigma^2, X) \tag{1}$$

$$= (\frac{1}{\sqrt{2\pi}\sigma})^n exp\{\frac{-1}{2\sigma^2}\sum_{i=1}^{n}(Y_i - X_i\mathbf{w})^2\} \tag{2}$$

$$= (2\pi\sigma^2)^{-N/2} exp\{\frac{-1}{2\sigma^2}(Y - X\mathbf{w})^T(Y - X\mathbf{w})\} \tag{3}$$

(b) Derive a closed-form solution for the Maximum Likelihood Estimate (**MLE**), in terms of the weight $\mathbf{w}$ and variance $\sigma^{\mathbf{2}}$.

*As can be seen, it's difficult to analyze the likelihood function as given by (3). Therefore, we take the* **Log Likelihood** *of it and then we have*

$$\mathcal{L}_{log} = -\frac{N}{2}ln(2\pi) - \frac{N}{2}ln(\sigma^2) - \frac{1}{2\sigma^2}(Y - X\mathbf{w})^T(Y - X\mathbf{w}) \tag{4}$$

*In (4), there are two parameters $\sigma^2$ and $\mathbf{w}$. In order to computer the optimal parameters such that $\mathcal{L}_{log}$ is maximum, we can compute the $\nabla\mathcal{L}_{log}$ in terms of $\sigma^2$ and $\mathbf{w}$.*

$$\frac{\partial\mathcal{L}_{log}}{\partial\mathbf{w}} = \frac{1}{\sigma^2}(Y - X\mathbf{w})^T X \tag{5}$$

$$\frac{\partial\mathcal{L}_{log}}{\partial\sigma^2} = -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4}(Y - X\mathbf{w})^T(Y - X\mathbf{w}) \tag{6}$$

*To find the maximum-likelihood estimator of $\mathbf{w}$, we set the derivative of equation (5) to zero. This gives a first-order condition for the maximisation of the log-likelihood function; and the solution of the equation is the optimal estimator*

$$\hat{\mathbf{w}} = (X^T X)^{-1} X^T Y \tag{7}$$

*Recalling what you have learned from lecture 3, you can find (7) is identical to the the estimator of $\mathbf{w}$ of linear regression. This can be seen from (4) that , if we fix $\sigma^2$ and compute the maximum of $\mathsf{L}_{log}$, the equation is equivalent as you compute for* **least square error** *of linear regression. which is*

$$argmax_{\mathbf{w}}\{-(Y - X\mathbf{w})^T(Y - X\mathbf{w})\}$$

*Actually, the identical form of $\mathbf{w}$ is not general, which is only true for the Gaussian $\epsilon$. Then to find the optimal estimator of $\sigma^2$, we set (6) to zero and substitute $\mathbf{w}$ by $\hat{\mathbf{w}}$, then we can solve*

$$\sigma^2(\mathbf{w}) = \frac{1}{N}(Y - X\hat{\mathbf{w}})^T(Y - X\hat{\mathbf{w}}). \tag{8}$$

□